

Uses, Costs and Specifications of Personal Computers

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Uses and benefits of an Office PC

Office PCs are generally quite low power, making them cheap to run. They can also be more reliable than high performance desktops, as they will be less likely to get hot. They can come in smaller sizes too, so they don't take up much space on a desk. The main uses of an Office PC are for document editing in the Office suite, communication via Teams or other messaging services, and general browsing, while some may be capable of more than this, it's not what they are built for and may run poorly.

Build costs and specs of an Office PC

I've chosen a [Office PC from Curry's](#) for demonstration which costs £380, you would be paying for an OEM build which could cost more than building a PC yourself, attempting to match the specs in PCPartPicker gives us a price of £465. The specs are weak for anything that isn't a basic task, anything that is resource intensive will more than likely run poorly, for example trying to run a modern resource intensive game on an Office PC like linked above, would either not run, or you would have to turn down graphic settings very far, which still might not be enough.

Uses and benefits of a Gaming PC

The main uses for a Gaming PC are for playing resource intensive games; however, it doesn't stop there as they also excel in performance at video editing in software like Premiere Pro, creating games in Unreal or Unity, or 3D modelling within Maya or Blender. The benefits are that they open many doors, as they can run almost anything you give it, which could be as simple as working on documents, or editing 8K long high frame rate videos without a stutter, and they are future proofed, they remain fast for many years.

Build costs and specs of a Gaming PC

I've chosen a [Gaming PC from Curry's](#) for demonstration which costs £4799, again you are paying for an OEM build which could cost more than building it yourself, matching the specs in PCPartPicker as close as possible gives us a price of £3776.

The specs in this PC are considerably stronger than the Office PC, it will be able to run almost any program you give it, many new resource intensive games will run extremely easy at the highest settings, 8K video editing in software like Premiere Pro at native resolution would run very well. Of course, it does not exclude it from doing basic tasks, which it will do without flaws.

Review of processor

Intel® Core™ Ultra 9 285K

Cores

24

Clock Speed

3.2GHz (4.6GHz Turbo)

Threads

24

Review

This processor will be able to easily handle any kind of task; it is as of writing the fastest Intel consumer focused processor available. This is thanks to the specifications Intel has packed into the processor, which it has 24 cores, 24 threads, and it can clock all the way up to 5.7GHz, with plenty of cache (36MB Intel Smart Cache and 40MB Total L2 Cache) to handle these operations

quickly. It is also able to utilise fast, large, and new RAM to its full potential, as it supports up to 192GB of total DDR5 RAM, at speeds of up to 6400MHz.

How can hardware components have effects on each other?

The hardware components you have put in your computer can have effects on how some or all of them will function, for example, if you chose a slow CPU and paired it with a flagship GPU you would most certainly create a bottleneck, where during intensive workloads your CPU would be working hard but your GPU would be barely able to even lift a finger, as the CPU would struggle to process the data fast enough for the GPU, which could also be incredibly inefficient for many reasons, the CPU could be drawing a lot of power for doing very little, and it would make all of the software on your computer extremely slow. This bottleneck could also reduce reliability overtime, as the CPU would be constantly working as hard as possible during intensive tasks, accelerating wear of the CPU, or you could experience data loss, prolonged intensive workloads could cause overheating, causing the system to force shutdown, which could corrupt system and user files and you would lose unsaved data.

Another example would be if you had the best CPU available for your motherboard, flagship GPU, best SSD available, etc, but that motherboard is from 2011, which would be inefficient, possibly unreliable and you would see slowdowns; especially during intensive tasks, as the motherboard may not be able to transfer the data around quick enough for the brand new installed components, as it may have older standards for data transfer that are too slow for the new components, which would again create a bottleneck, possibly across all components within the system. Older motherboards are also more prone to failure, as it's likely they've already had lots of wear, and older motherboards it was more common for them to fail unlike modern motherboards. Some also have less safety mechanisms in place to prevent data loss, such as critical hardware failure breaking multiple components, or USB overvoltage protection being missing.

These examples show that the specification of all components within your PC can affect the speed, reliability, and efficiency of your system and its software.

Some components can individually cause slowdowns on their own, like how using a slow HDD wouldn't cause the other components to struggle, they would be able to still do high workloads easily if they are built for it, however, getting software to start and begin fully running could take some time, as it would have to load from the slow HDD into memory.